Close faster, recover more

AI-powered CTV revenue reconciliation for Apex Exchange

Prepared for: CFO / VP Revenue Accounting · Director of Billing Operations · VP Data Engineering

1 · The headline

$18.1M of Q3 CTV revenue is in dispute with DSPs. $6.1M of it is recoverable, and today each case takes 3 weeks.

The Reconciliation Copilot finds every gap the day a partner's file lands, tells Billing Ops *why* it happened, drafts the dispute, and tracks it to cash, on one governed platform.

What changes	Today	With the Copilot
Root cause right on first look	82% (analyst rules of thumb)	**95%** (model, Jul–Sep test)
Time to resolve a dispute	21 days avg · 33 days p90	**Target ≤ 7 days**
Cases that need a credit or a reporting fix, not a chase	Not separated	**359 cases / $8.7M** routed straight to the right fix
Bad rows in partner files	Found in spreadsheets, if at all	**1,728 caught automatically** before billing

2 · Why this matters to Apex

- Apex sits in the middle of the money: **DSPs pay Apex, Apex pays publishers** and keeps its fee. When a DSP pays on a lower count than Apex owes the publisher, **Apex eats the difference**.
- Q3 2026: **$432M gross CTV revenue; $24.4M (5.7%) not matched by DSP reports.**
- 715 deal-months across **39 of 40 DSPs** needed review in Q3.
- Disputed invoices sit unpaid: about **$200K of disputed revenue a day**.

3 · Why the counts don’t match

Root cause	What happens in CTV	What to do	Q3 exposure
IVT filtering	The DSP’s fraud vendor removes spoofed CTV traffic	Credit only verified IVT	$8.2M
Billing error	Unexplained under-reporting on some days	Dispute (60% recovered historically)	$4.9M
Missing data	Days missing from the DSP’s report	Get the days, then bill (90% recovered)	$3.5M
Timezone shift	DSP reports in US time; primetime crosses midnight UTC	Ask for UTC reporting; no billing change	$0.6M
Within tolerance	Normal noise near the threshold	None	$0.9M

Recovery rates are Billing Ops’ actual Jan–Jun 2026 outcomes. Real cases are often mixed (e.g. a billing error plus some IVT); the model names the main cause.

4 · Today vs. tomorrow

Today: download each DSP's file → spreadsheet lookups → email back and forth → guess the cause → write off what's left.

Tomorrow: 1. Partner files are ingested and checked automatically (Lakeflow). 2. Every deal is reconciled against Apex's logs, daily and monthly. 3. A model explains *why* each gap happened, and a draft dispute is ready (AI). 4. Analysts work one ranked queue and resolve in a click (Databricks App + Lakebase). 5. Finance asks questions in plain English (Genie).

5 · The Reconciliation Workbench

- **Close overview:** September: Apex $153.5M vs DSP-reported $144.6M, **$8.9M at risk**, 219 cases, recovered-to-date.
- **Triage queue:** cases ranked by dollars at risk, with the model's root cause and confidence.
- **Case detail:** e.g. *Mosaic Bid · Bluebird TV OS · Entertainment PMP, September: $252,043 gap (23%) vs a 5% threshold. Model: billing error (99.8%).* Daily Apex-vs-DSP chart, evidence, AI case summary, an editable dispute email, and Send / Resolve / Approve credit.
- **Ask Genie:** *"What's our revenue at risk by partner team for September?"* → Central $3.52M, Atlantic $2.44M, Pacific $2.23M, EMEA $0.72M.

6 · Impact on your KPIs

KPI (owner)	Baseline (Q3 2026 data)	Target	Annual value*
Revenue recovered from DSPs (CFO)	$6.1M recoverable per quarter; team reaches ~60% of cases today*	Pursue every recoverable case	**≈ $10M / year**
Dispute cycle time / DSO (CFO)	21 days	≤ 7 days	**≈ $2.8M cash released** (one-time; ~$200K/day × 14 days)
Analyst effort (Billing Ops)	~270 cases/month × ~2.5 h* ≈ 4.2 FTE	~45 min per case (draft + evidence ready)	**≈ 3 FTE redeployed**
Right fix first time (Billing Ops)	Every flagged case gets chased	Half the queue goes straight to a credit or reporting fix	359 cases / quarter not chased
Month-end close (CFO)	~10 business days*	~4 days	Faster, auditable close

* Assumptions to validate in discovery: today's coverage (60%), manual effort per case (2.5 h), close length (10 days). Everything else is computed from the build's data.

7 · For the CFO

- **More cash, sooner:** ≈ $10M a year of recoverable revenue that currently goes stale, plus faster collection.
- **Audit-ready:** every number traces from the partner file to the invoice (Unity Catalog lineage), and every analyst action is logged.
- **Controls:** dollar figures are visible to Finance only; partner managers see only their own DSPs. Enforced by the platform, not the app.

8 · For the Director of Billing Operations

- One queue, ranked by dollars, so the team works the right cases first.
- The right root cause on first look 95% of the time, vs 82% today. Analysts confirm or override, and that feedback retrains the model.
- Dispute emails drafted from the evidence and the contract: 250 drafts, every dollar figure checked, every ask checked against the contract's billing basis, rated 4.8/5 for accuracy by an independent AI judge.
- Bad partner files caught on arrival, not at month-end.

9 · How it works: one governed platform

Need	Databricks
Ingest SSP logs + DSP reports, check quality	**Lakeflow**: bronze → silver → gold, 4 data-quality rules
Govern revenue data by partner and role	**Unity Catalog**: row filters, column masks, lineage
Explain the gap, draft the dispute	**ML** (MLflow, model registry) + **GenAI** (`ai_query`, Claude)
Live case workflow	**Lakebase** (Postgres), queryable live in Unity Catalog
Plain-English questions	**Genie**
The analyst's workbench	**Databricks App**

10 · Proof from the build

- 3.57M hourly log rows and 158K DSP report rows, covering 9 months, 40 DSPs and 800 CTV deals.
- 1,728 bad partner rows caught by the pipeline's quality rules.
- Root-cause model: F1 **0.92** on Jul–Sep vs **0.58** for today's rules (0.82 vs 0.56 against past analyst labels).
- GenAI: 100% of dollar figures grounded; 100% consistent with the contract; 0 parse failures.
- Genie: **12/12** finance questions answered correctly against reference SQL.
- App: tested end to end, from opening a case to sending the dispute to resolving with a credit. The result shows up in Unity Catalog and Genie immediately.

11 · Rollout plan

When	What	Success measure
Days 0–30	Pilot with the **top 5 DSPs (38% of Q3 revenue at risk)** on Apex's real files; analysts run the workbench alongside spreadsheets	Root-cause agreement with analysts ≥ 85%
Days 31–60	All 40 DSPs; on-behalf-of-user access in the app	Cycle time ≤ 10 days
Days 61–90	Retire the spreadsheet process; retrain on analyst-confirmed causes	Cycle time ≤ 7 days, close ≤ 4 days

12 · Next steps

1. **Discovery session with Billing Ops** to validate the assumptions (coverage, effort per case, close length).
2. **Data access:** one month of auction log extracts + DSP billing files for the top 5 DSPs.
3. **Pilot sponsor:** VP Revenue Accounting, with success criteria agreed up front.
4. **Executive readout** at day 30 with recovered dollars and cycle-time results.